$\boxed{\text{Step 7:}}$ Row 2 = Row 2 − 3×Row 3

$$\begin{bmatrix} 1 & 3 & 5 & 8 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\boxed{\text{Step 8:}}$ Row 1 = Row 1 − 3×Row 2

$$\begin{bmatrix} 1 & 0 & -1 & 8 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We will now show that this matrix in Step 8 is in rref form.

1. There are no zero rows in this matrix. ∴ The condition that all the zero rows must be below all non-zero rows is vacuously true.

2. The leading entry of row 1 = 1 in Column 1
   The leading entry of row 2 = 1 in Column 2
   The leading entry of row 3 = 1 in Column 4

The condition that the leading entry of a row has a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
   Row 3 Column 2 = 0
   ∴ The condition that for each leading entry of a row, the columns elements below them are all zero's is true.

4. From 2, we Know all leading entries are 1. ~~We can now conclude that each~~

5. Row 1 Column 2 = 0. ~~From 3, we can~~
   Row 1 Column 4 = Row 1 Column 4 = 0
   Row 2 Column 4 = 0

From 3, we can now conclude that each leading 1 is the only non-zero element in its respective column

∴ The matrix $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is in rref form □